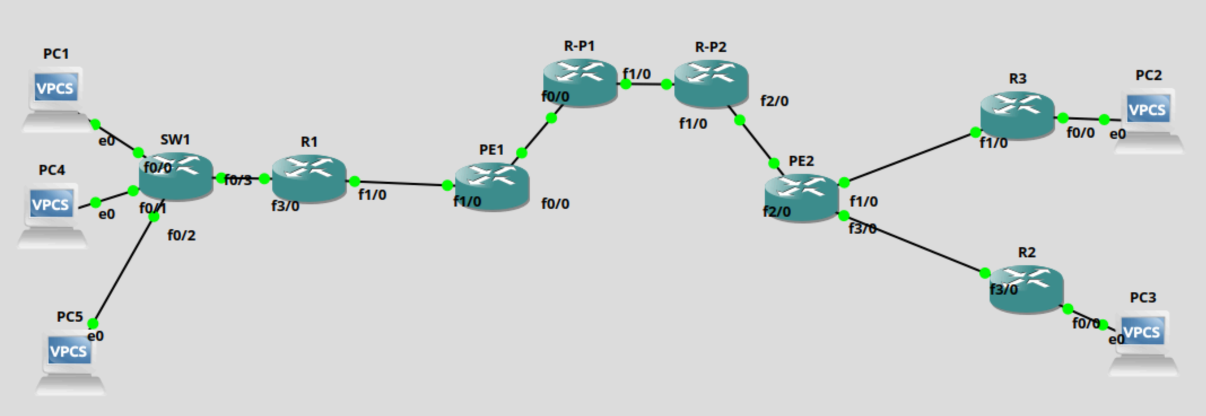
TP1 - R4.ROM10

Voici le plan de notre maquette :



R1 :

```
en
conf t
! Configuration des sous-interfaces (VLANs)
int f3/0
no sh
int f3/0.100
encapsulation dot1Q 100
ip address 192.168.1.254 255.255.255.0
! Application des ACLs (supposons PC4 = .4 et PC1 = .1)
ip access-group 100 in
ip access-group 101 out
int f3/0.101
encapsulation dot1Q 101
ip address 172.16.1.254 255.255.255.0

! Interco vers PE1 (Sous-réseau 1)
int f1/0
ip address 10.151.151.2 255.255.255.252
no sh

! Routage eBGP vers PE1
```

```
router bgp 65102
neighbor 10.151.151.1 remote-as 65101
network 192.168.1.0 mask 255.255.255.0
network 172.16.1.0 mask 255.255.255.0

! Serveur DHCP global
ip dhcp excluded-address 192.168.1.254
ip dhcp excluded-address 172.16.1.254
ip dhcp excluded-address 192.168.2.254
ip dhcp excluded-address 192.168.3.254

ip dhcp pool VLAN100
network 192.168.1.0 255.255.255.0
default-router 192.168.1.254

ip dhcp pool VLAN101
network 172.16.1.0 255.255.255.0
default-router 172.16.1.254

ip dhcp pool LAN_R2
network 192.168.2.0 255.255.255.0
default-router 192.168.2.254

ip dhcp pool LAN_R3
network 192.168.3.0 255.255.255.0
default-router 192.168.3.254

! ACLs pour empêcher PC4 de joindre PC1, tout en permettant
à PC1 de joindre PC4
access-list 100 permit ip host 192.168.1.4 host 192.168.1.1
access-list 100 deny ip host 192.168.1.4 any
access-list 100 permit ip any any

access-list 101 permit ip host 192.168.1.1 host 192.168.1.4
access-list 101 deny ip any host 192.168.1.4
access-list 101 permit ip any any
exit
```

PE1 :

```
en
conf t
! Création de la VRF CUST
ip vrf CUST
rd 10:100
route-target export 100:1000
route-target import 100:1000

! Interface Loopback
int lo0
ip address 1.1.1.1 255.255.255.255
no sh

! Interco Cœur vers R-P1 (Sous-réseau 4)
mpls label range 100 199
int f0/0
ip address 10.151.151.13 255.255.255.252
mpls ip
no sh

! Interco CE vers R1 (Sous-réseau 1)
int f1/0
ip vrf forwarding CUST
ip address 10.151.151.1 255.255.255.252
no sh

! OSPF pour le cœur MPLS
router ospf 1
network 1.1.1.1 0.0.0.0 area 0
network 10.151.151.12 0.0.0.3 area 0

! MP-BGP
router bgp 65101
! iBGP (VPNv4) vers PE2
neighbor 4.4.4.4 remote-as 65101
neighbor 4.4.4.4 update-source lo0
```

```
address-family vpnv4
neighbor 4.4.4.4 activate
neighbor 4.4.4.4 send-community extended
exit-address-family

! eBGP (IPv4) vers R1 dans la VRF
address-family ipv4 vrf CUST
neighbor 10.151.151.2 remote-as 65102
neighbor 10.151.151.2 activate
exit-address-family
exit
```

R-P1 :

```
en
conf t
mpls label range 200 299

int lo0
ip address 2.2.2.2 255.255.255.255
no sh

! Vers PE1 (Sous-réseau 4)
int f0/0
ip address 10.151.151.14 255.255.255.252
mpls ip
no sh

! Vers R-P2 (Sous-réseau 5)
int f1/0
ip address 10.151.151.17 255.255.255.252
mpls ip
no sh

! OSPF Cœur
router ospf 1
network 2.2.2.2 0.0.0.0 area 0
network 10.151.151.12 0.0.0.3 area 0
```

```
network 10.151.151.16 0.0.0.3 area 0
exit
```

R-P2 :

```
en
conf t
mpls label range 300 399

int lo0
ip address 3.3.3.3 255.255.255.255
no sh

! Vers R-P1 (Sous-réseau 5)
int f1/0
ip address 10.151.151.18 255.255.255.252
mpls ip
no sh

! Vers PE2 (Sous-réseau 6)
int f2/0
ip address 10.151.151.21 255.255.255.252
mpls ip
no sh

! OSPF Cœur
router ospf 1
network 3.3.3.3 0.0.0.0 area 0
network 10.151.151.16 0.0.0.3 area 0
network 10.151.151.20 0.0.0.3 area 0
exit
```

PE2 :

```
en
conf t
! Création de la VRF CUST
ip vrf CUST
```

```
rd 10:100
route-target export 100:1000
route-target import 100:1000

! Interface Loopback
int lo0
ip address 4.4.4.4 255.255.255.255
no sh

! Interco Cœur vers R-P2 (Sous-réseau 6)
mpls label range 400 499
int f2/0
ip address 10.151.151.22 255.255.255.252
mpls ip
no sh

! Interco CE vers R2 (Sous-réseau 2)
int f3/0
ip vrf forwarding CUST
ip address 10.151.151.5 255.255.255.252
no sh

! Interco CE vers R3 (Sous-réseau 3)
int f1/0
ip vrf forwarding CUST
ip address 10.151.151.9 255.255.255.252
no sh

! OSPF pour le cœur MPLS
router ospf 1
network 4.4.4.4 0.0.0.0 area 0
network 10.151.151.20 0.0.0.3 area 0

! Route Statique VRF pour joindre le LAN de R3
ip route vrf CUST 192.168.3.0 255.255.255.0 10.151.151.10

! MP-BGP
router bgp 65101
```

```
! iBGP (VPNv4) vers PE1
neighbor 1.1.1.1 remote-as 65101
neighbor 1.1.1.1 update-source lo0
address-family vpnv4
neighbor 1.1.1.1 activate
neighbor 1.1.1.1 send-community extended
exit-address-family

! eBGP vers R2 + Redistribution de la route statique (pour
annoncer le LAN R3 à R1)
address-family ipv4 vrf CUST
neighbor 10.151.151.6 remote-as 65103
neighbor 10.151.151.6 activate
redistribute static
exit-address-family
exit
```

R2 :

```
en
conf t
! Interco vers PE2 (Sous-réseau 2)
int f3/0
ip address 10.151.151.6 255.255.255.252
no sh

! Interface LAN (PC3) avec relais DHCP
int f0/0
ip address 192.168.2.254 255.255.255.0
ip helper-address 10.151.151.2
no sh

! Routage eBGP vers PE2
router bgp 65103
neighbor 10.151.151.5 remote-as 65101
network 192.168.2.0 mask 255.255.255.0
exit
```

R3 :

```
en
conf t
! Interco vers PE2 (Sous-réseau 3)
int f1/0
ip address 10.151.151.10 255.255.255.252
no sh

! Interface LAN (PC2) avec relais DHCP
int f0/0
ip address 192.168.3.254 255.255.255.0
ip helper-address 10.151.151.2
no sh

! Routage statique par défaut vers PE2
ip route 0.0.0.0 0.0.0.0 10.151.151.9
exit
```

Vérifications :

PC1 :

```
PC1> ip dhcp
DORA IP 192.168.1.1/24 GW 192.168.1.254
```

PC5 :

```
PC5> ip dhcp
DDORA IP 172.16.1.1/24 GW 172.16.1.254
```

PC4 :

```
PC4> ip 192.168.1.10 /24 192.168.1.254
Checking for duplicate address...
PC4 : 192.168.1.10 255.255.255.0 gateway 192.168.1.254
```

PC2 et PC3 :

Je n'ai pas réussi a joindre le DHCP

R1 :

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/24 is subnetted, 1 subnets
C       172.16.1.0 is directly connected, FastEthernet3/0.101
     10.0.0.0/30 is subnetted, 1 subnets
C       10.151.151.0 is directly connected, FastEthernet1/0
C    192.168.1.0/24 is directly connected, FastEthernet3/0.100
B    192.168.2.0/24 [20/0] via 10.151.151.1, 00:02:16
B    192.168.3.0/24 [20/0] via 10.151.151.1, 00:02:16
```

PE1 :

```
PE1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     1.0.0.0/32 is subnetted, 1 subnets
C       1.1.1.1 is directly connected, Loopback0
     2.0.0.0/32 is subnetted, 1 subnets
O       2.2.2.2 [110/2] via 10.151.151.14, 00:03:27, FastEthernet0/0
     3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/3] via 10.151.151.14, 00:03:27, FastEthernet0/0
     4.0.0.0/32 is subnetted, 1 subnets
O       4.4.4.4 [110/4] via 10.151.151.14, 00:03:27, FastEthernet0/0
     10.0.0.0/30 is subnetted, 3 subnets
C       10.151.151.12 is directly connected, FastEthernet0/0
O       10.151.151.16 [110/2] via 10.151.151.14, 00:03:28, FastEthernet0/0
O       10.151.151.20 [110/3] via 10.151.151.14, 00:03:28, FastEthernet0/0
```

```
PE1#show ip route vrf CUST

Routing Table: CUST
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/24 is subnetted, 1 subnets
B       172.16.1.0 [20/0] via 10.151.151.2, 00:03:27
     10.0.0.0/30 is subnetted, 1 subnets
C       10.151.151.0 is directly connected, FastEthernet1/0
B    192.168.1.0/24 [20/0] via 10.151.151.2, 00:03:27
B    192.168.2.0/24 [200/0] via 4.4.4.4, 00:03:18
B    192.168.3.0/24 [200/0] via 4.4.4.4, 00:03:18
```

```
PE1#traceroute 4.4.4.4

Type escape sequence to abort.
Tracing the route to 4.4.4.4

  1 10.151.151.14 [MPLS: Label 200 Exp 0] 40 msec 32 msec 32 msec
  2 10.151.151.18 [MPLS: Label 300 Exp 0] 28 msec 32 msec 20 msec
  3 10.151.151.22 28 msec 32 msec 32 msec
```

PE2 :

```
PE2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     1.0.0.0/32 is subnetted, 1 subnets
O       1.1.1.1 [110/4] via 10.151.151.21, 00:04:42, FastEthernet2/0
     2.0.0.0/32 is subnetted, 1 subnets
O       2.2.2.2 [110/3] via 10.151.151.21, 00:04:42, FastEthernet2/0
     3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/2] via 10.151.151.21, 00:04:42, FastEthernet2/0
     4.0.0.0/32 is subnetted, 1 subnets
C       4.4.4.4 is directly connected, Loopback0
     10.0.0.0/30 is subnetted, 3 subnets
O       10.151.151.12 [110/3] via 10.151.151.21, 00:04:43, FastEthernet2/0
O       10.151.151.16 [110/2] via 10.151.151.21, 00:04:43, FastEthernet2/0
C       10.151.151.20 is directly connected, FastEthernet2/0
```

```
PE2#show ip route vrf CUST

Routing Table: CUST
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/24 is subnetted, 1 subnets
B       172.16.1.0 [200/0] via 1.1.1.1, 00:04:26
     10.0.0.0/30 is subnetted, 3 subnets
C       10.151.151.8 is directly connected, FastEthernet1/0
B       10.151.151.0 [200/0] via 1.1.1.1, 00:04:26
C       10.151.151.4 is directly connected, FastEthernet3/0
B    192.168.1.0/24 [200/0] via 1.1.1.1, 00:04:42
B    192.168.2.0/24 [20/0] via 10.151.151.6, 00:28:43
S    192.168.3.0/24 [1/0] via 10.151.151.10
```

R2 :

```
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/24 is subnetted, 1 subnets
B       172.16.1.0 [20/0] via 10.151.151.5, 00:05:27
     10.0.0.0/30 is subnetted, 2 subnets
B       10.151.151.0 [20/0] via 10.151.151.5, 00:05:27
C       10.151.151.4 is directly connected, FastEthernet3/0
B    192.168.1.0/24 [20/0] via 10.151.151.5, 00:05:27
C    192.168.2.0/24 is directly connected, FastEthernet0/0
B    192.168.3.0/24 [20/0] via 10.151.151.5, 00:28:17
```

R3 :

```
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.151.151.9 to network 0.0.0.0

     10.0.0.0/30 is subnetted, 1 subnets
C       10.151.151.8 is directly connected, FastEthernet1/0
C    192.168.3.0/24 is directly connected, FastEthernet0/0
S*   0.0.0.0/0 [1/0] via 10.151.151.9
```

Fonctionnement DHCP :

Le DHCP fonctionne selon le principe DORA,

La machine envoie un Discover, pour demander a un dhcp une adresse,

Elle reçoit une Offer, Elle accepte avec le Request et elle Acknowledge qu’elle utilise bien cette ip là.

Fonctionnement MPLS :

Le MPLS permet de transmettre du traffic sous forme de packet encapsulé avec un tag qui spécifie un router spécifique, ce protocole est rapide, il isole les packets et les sécurise mais il est difficile a mettre en place et extrèmement couteux.